

Session 4: Reproducible Documents

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1 Session4

Today, we are going to learn how to design algorithm, including: * What is Reproducible Documents and how to create/publish them with quarto * Write metadata for Reproducible Documents (above cell) * How to create array with different numpy attributes and method * Design an algorithm

Note: This notebook is copied and modified from session3

1.1 Example content of Reproducible document

In this example, formula can be easily handled by Markdown

Rocket Equation

At time t_0 to rocket starts to expel gas at a constant mass flow rate R measured in kg/s and exhaust velocity relative to the rocket v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \quad (1)$$

Integrating both sides of Equation 1 from 0 to T we get

1.2 Open the file

/home/shch14/spaceMed2026/shch14

1.3 Extract data

63065 63065

```
25.34206948640483 1388.0
25.344574881312038 1388.0
25.347080276219245 1385.0
25.349585671126455 1383.0
25.352091066033662 1383.0
25.35459646094087 1378.0
25.357101855848075 1378.0
25.359607250755285 1375.0
25.362112645662492 1375.0
25.3646180405697 1373.0
```

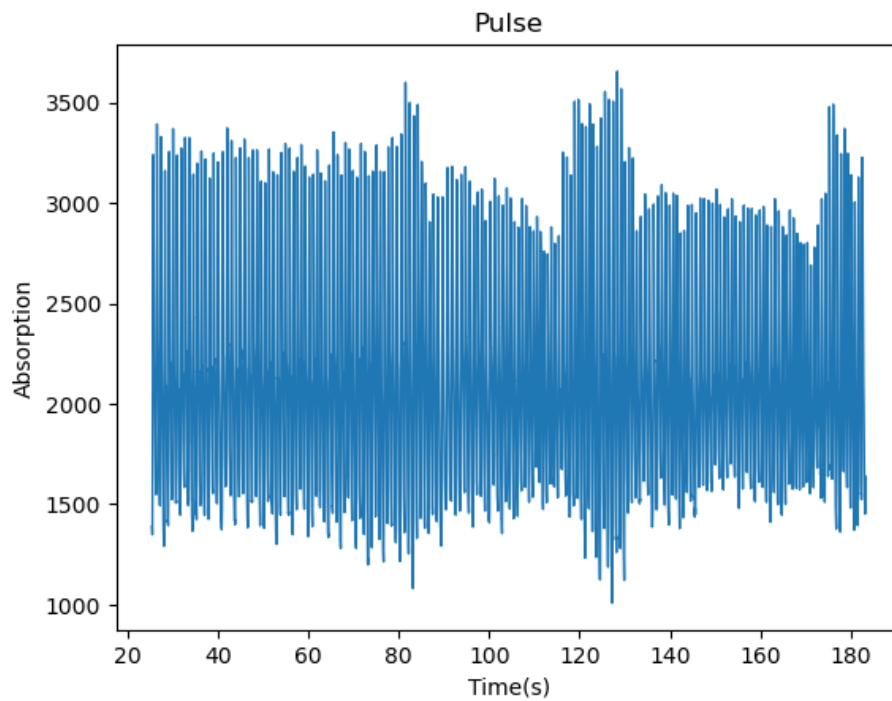


Figure 1: Measurements from an oxymetry sensor as a function of time for 3 cycles

1.4 Python Array

```
[1 2 3 4 5]
```

```
(5,)
```

```
dtype('int64')
```

```
40
```

```
[1. 2. 3. 4. 5.]
```

```

[[1 2]
 [3 4]]

[[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]]

(10, 20)

[ 1  2  3  4  5  6  7  8  9 10]

(10,)

dtype('int64')

80

np.int64(385)

[1.e+01 1.e+02 1.e+03 1.e+04 1.e+05 1.e+06 1.e+07 1.e+08 1.e+09 1.e+10]

(10,)

(2, 3)

array([[ 2,  8],
       [ 4, 10],
       [ 6, 12]])

array([[17, 22, 27],
       [22, 29, 36],
       [27, 36, 45]])

[ 0.   0.5  1.   1.5  2.   2.5  3.   3.5  4.   4.5  5.   5.5  6.   6.5
  7.   7.5  8.   8.5  9.   9.5 10.]

(21,)

array([1., 2., 3., 4., 5.])

np.float64(5.0)

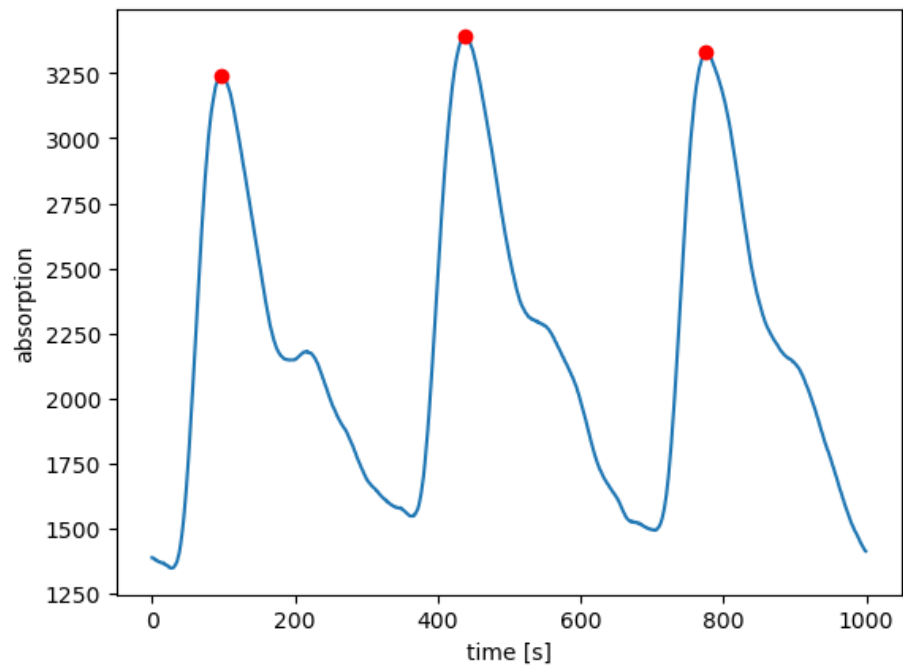
np.int64(4)

```

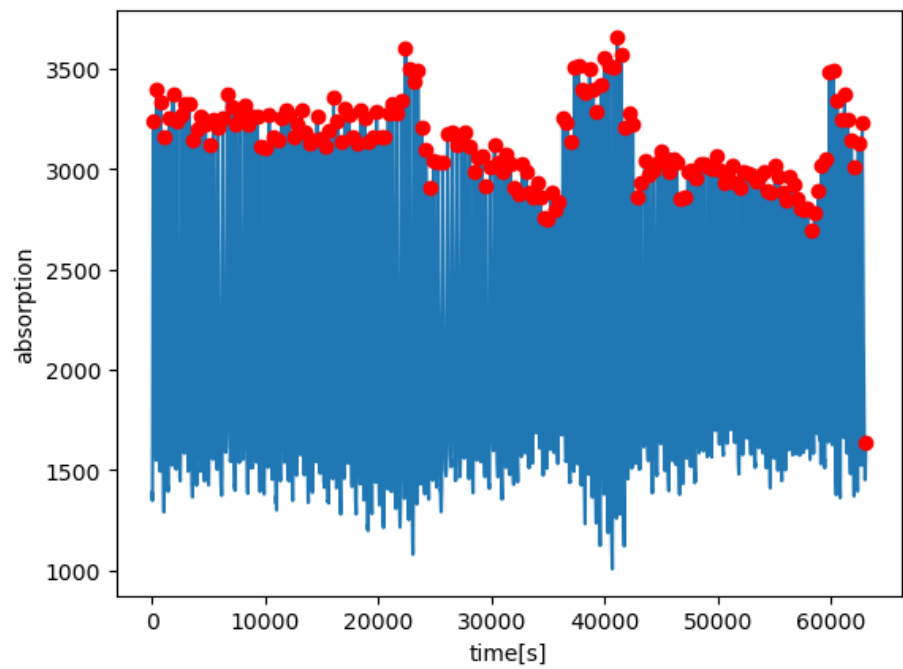
Exercise - Find the peaks

- Consider single cycle and come up with a naive solution
- store all peak locations in a list or array
- compute heart rate and plot it

```
[96, 438, 774]
```



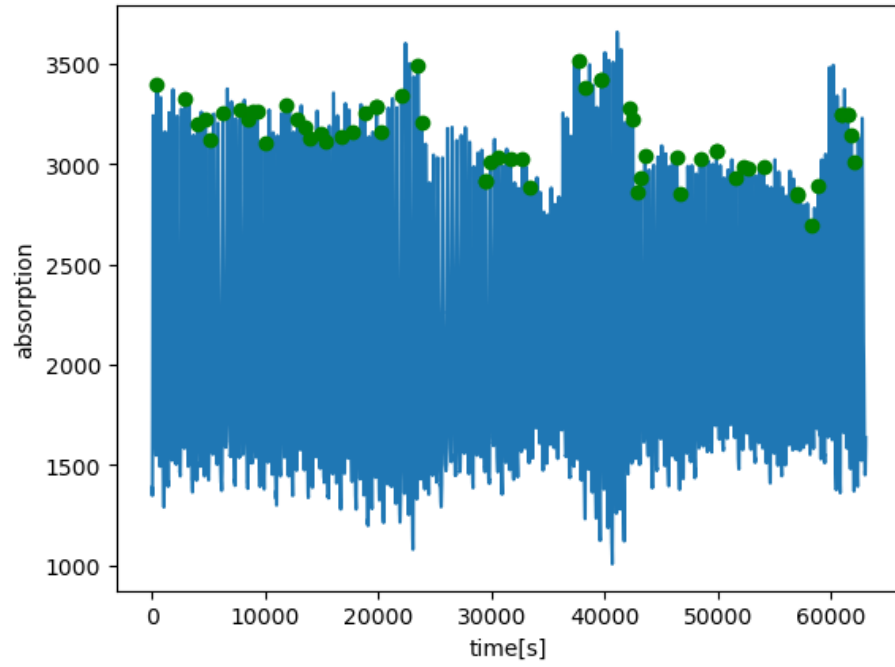
[96, 438, 774, 1142, 1502, 1851, 2211, 2578, 2929, 3291, 3651, 4003, 4356, 4724, 5092, 5451]



The logic we discussed before instruction

check the locations where signal transitions from positive to negative, and minimize noise by using a defined amplitude threshold

[96, 438, 774, 1142, 1502, 1851, 2211, 2578, 2929, 3291, 3651, 4003, 4356, 4724, 5092, 545



1.5 HR Calculation

